SIMATS ENGINEERING

Department of Information Technology

ITA05 – COMPUTER VISION

ASSIGNMENT 1 – CO1

Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Vision-Based PCB Solder Joint Inspection Line

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

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Task 1 – Inspection Approach Comparison

Approach A – Manual Inspection

Manual inspection uses trained operators and a magnifier lamp. It has low initial equipment cost and can recognize unusual visual conditions when an experienced inspector understands the board. Human judgement is flexible and can use context that has not been encoded in software. However, fatigue, attention changes, visual acuity and differences in training cause inconsistent decisions.

At two boards per second, a ten-hour shift contains 72,000 boards. One inspector cannot inspect every board thoroughly at this rate. Manual inspection can remain useful for sampling, exception review and confirmation of newly observed defect types, but it cannot be the primary full-rate inspection method for this conveyor.

An appropriate manual study would use three evaluators, two rounds and a common defect reference sheet. Record PASS/REJECT decisions separately for every board. Cohen's kappa can quantify agreement between two evaluators; weighted or multi-rater statistics may be needed when more than two evaluators are analysed. Do not claim measured kappa values until the actual evaluation sheet is completed.

Task 1 – Inspection Approach Comparison

Approach B – Image-Processing-Assisted Inspection

In the assisted approach, a camera captures each PCB and software improves the display using operations such as cv2.equalizeHist, cv2.resize and digital zoom. The operator still makes the final decision. Enhancement can make low-contrast regions easier to see, but it cannot recover information that was never captured because of blur, saturation or insufficient spatial sampling.

Image enhancement must be applied consistently. Histogram equalisation can improve global contrast but may also amplify noise or make solder glare appear more prominent. A displayed image should preserve access to the original image so the operator can compare enhanced and unmodified data. Every enhancement setting should be recorded for traceability.

This approach reduces the effort of searching a full board and may improve visual consistency, but the operator remains a throughput bottleneck. It is best positioned as a review station for borderline cases, new board variants and audit investigations rather than the main reject decision.

Task 1 – Inspection Approach Comparison

Approach C – Automated Computer Vision

An automated system captures an image, locates the PCB, aligns it to a reference, segments components and solder regions, extracts measurements and applies rules or a classifier. It then sends a timed accept/reject signal to the gate. The system can inspect continuously and produce repeatable measurements when camera calibration, lighting and board positioning are controlled.

Automation does not mean that every decision is infallible. Rules may miss defects that were not anticipated, while a classifier may fail under domain shift, new component finishes or unusual reflections. The design therefore needs a reject-safe policy, periodic validation, retained images, a controlled recipe for each board revision and a manual review process for uncertain cases.

Recommended architecture: use automated vision for every board, preserve the raw image, store extracted measurements and rule evidence, and route low-confidence or unfamiliar conditions to a human review queue. A known defect should never be silently converted to PASS merely because a software module fails.

Task 1 – Inspection Approach Comparison

Comparison and Evaluation Method

Compare the three approaches using the same labelled board set. The brief requires at least 30 board images, including at least 10 with seeded defects. For each board, record defect truth, decision, processing or inspection time and operator identity where relevant. Split the dataset by physical board or capture session so near-identical frames do not leak between training and test sets.

Useful metrics include defect recall, false-accept rate, false-reject rate, precision, repeatability, throughput and latency. Escaped-defect rate is especially important because the customer allows at most 1% escaped defective boards. A high overall accuracy can hide poor performance on rare defects, so report per-defect recall and a confusion matrix.

Timing should be measured from image acquisition to completed decision, not merely the time taken by one OpenCV function. Record median and high-percentile latency. The 1.5-second limit is a system requirement, while the physical travel time from camera to gate is 0.6 m / 0.4 m/s = 1.5 seconds. The controller must therefore account for communication and actuator delay as well as image processing.

Task 1 – Inspection Approach Comparison

Statistical Agreement and Throughput

For two raters, Cohen's kappa is $\kappa = (Po - Pe) / (1 - Pe)$, where Po is observed agreement and Pe is agreement expected by chance from the raters' marginal proportions. A value near 1 indicates strong agreement; a value near 0 indicates chance-level agreement. Report the underlying contingency table because kappa can be affected by class imbalance.

Intra-rater agreement is measured by comparing the same inspector's first and second round. Inter-rater agreement compares different inspectors. For automated systems, repeat the same image under identical settings to test deterministic repeatability, then repeat physical captures to include acquisition variation.

At 2 boards/s, 7,200 boards pass per hour and 72,000 pass in ten hours. This is the required line volume, not a realistic manual capacity estimate. Measure manual inspection capacity in a timed trial and calculate the shortfall. The difference demonstrates why human inspection should be used for exception handling rather than 100% inspection.

Task 1 – Inspection Approach Comparison

Recommendation and Limits

Approach C is the recommended primary inspection method because it can operate at the required line rate, measure placement offsets, create traceable records and trigger the reject gate consistently. Approach B is recommended as a human review interface. Approach A remains useful for establishing ground truth, reviewing difficult examples and auditing automated decisions.

The automated system should be accepted only after testing against the full defect catalogue and the 1% escape requirement. A defect type absent from the rule set may be passed if it does not violate any existing rule. Corrective measures include collecting the new defect examples, updating the defect taxonomy, adding a detection rule or retraining the classifier, and validating the change on an independent test set.

A vision-language model may provide useful descriptions but should not be the sole customer-audited decision-maker without repeatability, latency and reliability validation. Model version, prompt, image preprocessing and response must be logged. A fixed prompt on ten images is only a preliminary study, not evidence that the model meets a 1% escape limit.

Inspection method	Strength	Main limitation	Recommended role
Manual	Flexible human judgement	Fatigue and low line capacity	Audit and exception review
Assisted	Enhanced image for operator	Human remains decision bottleneck	Review station
Automated	Repeatable measurements and gate signal	Needs validation and maintenance	Primary inspection

Task 2 – Pipeline Design and Operation

Three Levels of Vision

Low-level vision transforms pixels into improved pixels. Examples in this assignment include Gaussian blur, image subtraction against a golden board and thresholding. Mid-level vision converts images into regions and measurable attributes such as component bounding boxes, centroids, blob areas and solder-mask regions. High-level vision interprets those attributes and produces a decision such as PASS, REWORK or SCRAP.

The three candidate pipelines differ in how far they interpret the image. Pipeline A stops at pixel differences. Pipeline B adds alignment, segmentation and feature extraction. Pipeline C adds a classifier or decision stage to interpret feature vectors. The deeper pipeline can be more tolerant of irrelevant pixel variation, provided its upstream alignment and features are correct.

Each stage should define its input and output explicitly. For example, image registration takes a captured image and reference image and returns a transform; segmentation takes an aligned image and returns binary masks; feature extraction returns a vector; classification returns a class and confidence.

Task 2 – Pipeline Design and Operation

Pipeline A – Low-Level Only

Pipeline A applies Gaussian blur to reduce sensor noise, computes an absolute difference from a golden image and thresholds the result. A fixed count of differing pixels is used to classify the board. It is computationally simple and easy to implement with OpenCV.

The weakness is sensitivity to nuisance variation. A board shifted by one millimetre may produce a large difference image even when every component is correct. A shadow from a 12 mm capacitor, a silkscreen revision or a small exposure change can also create many differing pixels. A threshold cannot reliably distinguish these causes from solder defects.

Low-level comparison is useful for detecting gross changes when the board is accurately registered and lighting is stable. It should not be used alone for precision placement measurement or defect classification.

Task 2 – Pipeline Design and Operation

Pipeline B – Low- and Mid-Level

Pipeline B first aligns the captured PCB with a golden reference. Template matching or feature-based registration can estimate translation and rotation. Contours and connected components then identify candidate components and solder regions. Bounding rectangles, centroids, areas, aspect ratios and distances to reference coordinates form the feature set.

Mid-level features reduce sensitivity to small pixel changes because the decision is based on object geometry and region properties. For example, the centroid of a component can be compared with its nominal centre after calibration. A missing component can be detected when the expected component region contains no valid component blob.

This pipeline is more explainable than a purely learned classifier. Its limits are segmentation errors, overlapping components, reflective solder and board variants that change the expected geometry. It needs carefully designed thresholds and a board-specific reference recipe.

Task 2 – Pipeline Design and Operation

Pipeline C – Low, Mid and High-Level

Pipeline C extends Pipeline B by passing extracted feature vectors to a classifier such as an SVM or Random Trees model. Inputs may include component presence, placement offset, solder-region area, local brightness, colour statistics, bridge candidate geometry and texture measures. The output can be PASS, REWORK or SCRAP with a confidence score.

A classifier can combine several weak indicators and may generalize better than a long list of independent thresholds. It does not automatically solve poor image quality. If alignment is wrong or the segmentation mask excludes part of a solder joint, the classifier receives incorrect evidence and may make a confident error.

Use a separate validation set to choose operating thresholds. For a safety-oriented inspection line, a low-confidence or out-of-distribution sample should be rejected for review rather than automatically accepted.

Task 2 – Pipeline Design and Operation

Evaluation Dataset and Metrics

All three pipelines must be tested on the same labelled images. Include nominal boards, solder bridges, solder balls, missing components, insufficient solder, board shifts, silkscreen differences and shadows from tall capacitors. Keep defect labels at both board level and region level so that missed defects can be identified.

Report accuracy, precision, recall, false-accept rate, false-reject rate and latency in milliseconds per image. Also report performance on each challenging condition. A single aggregate accuracy number can conceal the fact that a pipeline works well on clean images but fails on shifted boards.

Measure processing time using a monotonic timer around the complete pipeline. Run warm-up iterations and report the median and 95th percentile. The decision requirement is 1.5 seconds; the system must also meet the sustained arrival rate of two boards per second without an ever-growing queue.

Task 2 – Pipeline Design and Operation

Selected Pipeline and Stage Annotation

The recommended design is Pipeline C with deterministic low- and mid-level preprocessing followed by a compact, explainable classifier. Low-level stages: colour conversion, denoising, flat-field correction and thresholding. Mid-level stages: board alignment, component ROI extraction, connected-component labelling and feature calculation. High-level stage: class interpretation and final routing decision.

Example representation sequence: BGR image → corrected image → aligned board image → component and solder masks → feature vector → defect class and confidence → PASS/REJECT. The pipeline should save intermediate masks for a sample of boards and for every rejected or uncertain board.

Keep hard safety rules for clearly defined defects such as a bridge between adjacent pads. A learned model can assist with subtle insufficient-solder appearance, but its output should be constrained by calibrated geometry and validated decision thresholds.

Task 2 – Pipeline Design and Operation

Deep Models and Final Choice

CNNs learn local filters and progressively larger visual patterns. Vision Transformers use attention to model relationships among image patches. Autoencoders learn a representation of normal data and may flag unusual regions; PatchCore compares local feature embeddings with a memory bank of normal examples. These methods may help identify previously unseen appearance changes.

Deep models generally require representative training data, careful validation and more deployment effort than traditional OpenCV rules. The brief specifies a four-core CPU and no GPU, so a compact classical pipeline is a practical first implementation. A pretrained model may be evaluated offline, but latency must be measured on the actual edge PC before deployment.

Final choice: Pipeline C, with rule-based geometric checks and a modest classifier, is the strongest balance of explainability, tolerance and implementation cost. It remains dependent on reliable image acquisition, representative labels and ongoing validation whenever board designs change.

Pipeline	Stages	Strength	Main risk
A: Low	Blur, absdiff, threshold	Simple and fast	Shift/shadow false alarms
B: Low + mid	Alignment, segmentation, features	Geometric interpretation	Segmentation error
C: Three-level	Features + classifier	Combines multiple cues	Needs labelled validation

Task 3 – Image Representation Analysis

Digital Image and PCB Colour Information

A digital image is represented as an array f(x,y). A grayscale image has one intensity value per pixel, while a colour image has three channels. For OpenCV, a standard colour image is usually stored in BGR channel order. The selected representation determines which visual differences are available to the inspection algorithm.

PCB inspection must separate green solder mask, metallic solder, copper or gold pads, component bodies and flux discolouration. Brightness alone may not distinguish these materials under changing illumination. Colour spaces that separate chromaticity from brightness can improve segmentation and make the system less sensitive to exposure changes.

The three representations compared here are grayscale, HSV and CIE L*a*b*. Each should be applied to the same set of at least 30 solder-joint samples under two lighting levels, as requested in the question.

Task 3 – Image Representation Analysis

A – Grayscale Representation

Convert BGR to grayscale with cv2.cvtColor(img, cv2.COLOR_BGR2GRAY). The resulting image has one 8-bit channel with values from 0 to 255. This representation is compact and computationally inexpensive. Brightness thresholds can identify very dark or bright regions when the illumination and surface finish are controlled.

Grayscale discards hue and saturation. A brownish flux stain and a dark shadow may have similar intensity, while silver solder and a bright component marking may both appear bright. If overhead illumination dims by half, the intensity distribution shifts and a threshold selected earlier may no longer separate good and insufficient solder.

Grayscale is useful for edge detection, shape analysis and local contrast measurements, but it should not be the only signal for solder quality when colour is part of the defect definition.

Task 3 – Image Representation Analysis

B – HSV Representation

HSV represents hue, saturation and value. Hue describes the dominant colour, saturation describes colourfulness and value represents brightness. In OpenCV, hue is stored on a scaled range from 0 to 179 for 8-bit images, while saturation and value range from 0 to 255.

HSV can help separate the green solder mask from metallic and copper-coloured regions. Saturation and value thresholds can be combined to suppress some glare, although specular highlights may become nearly white and have low saturation. Hue is circular, so values near 0 and 179 are adjacent rather than far apart; thresholding across the wrap boundary requires two ranges or a circular distance.

HSV is a practical choice for rule-based colour segmentation, but it is not fully invariant to illumination. It should be used with controlled lighting, exposure locking and region-specific thresholds.

Task 3 – Image Representation Analysis

C – CIE L*a*b* Representation

Lab separates lightness L* from two colour-opponent channels a* and b*. The a* axis represents green-to-red variation and the b* axis represents blue-to-yellow variation. This makes Lab useful for detecting flux discolouration and comparing a solder region with a reference colour.

Colour difference can be calculated using ΔE. A small ΔE indicates similar colours, while a larger value indicates a greater colour difference. The chosen ΔE formula and colour-management assumptions must be stated. OpenCV's 8-bit Lab encoding is scaled and offset, so raw channel values should not be interpreted as physical CIELAB units without conversion.

Lab is useful when colour differences are subtle and customer explanation matters. It is more computationally involved than grayscale but remains practical for small regions of interest.

Task 3 – Image Representation Analysis

Comparison Under Illumination and Glare

Compare the representations on solder, mask and pad regions. Measure class separation using channel histograms or scatter plots. Repeat captures at two controlled lighting levels and report overlap between good and defective joint distributions. Segmentation accuracy should be calculated against manually verified masks.

Grayscale is most sensitive to a global brightness shift. HSV provides a separate hue channel but may be unstable for low-saturation metallic pixels and near the hue wrap boundary. Lab separates lightness and chromaticity and is often easier to interpret for colour difference, but specular reflections still alter the observed surface appearance.

Glare is a photometric effect, not simply a colour-space problem. A saturated highlight can destroy texture and colour evidence in every representation. Diffuse or dome lighting, cross-polarization where appropriate, locked exposure and highlight masks are therefore required.

Task 3 – Image Representation Analysis

Array Structure, Storage and Arithmetic

For an image of M rows, N columns and k channels, the raw storage requirement is M × N × k × b bits, where b is the bits per channel. For an 8-bit 2448 × 2048 BGR image, storage is 2448 × 2048 × 3 bytes = 15,040,512 bytes, approximately 15.04 MB in decimal units before file compression or metadata.

In NumPy, an image is indexed as image[row, column, channel], whereas geometric coordinates are commonly written as (x,y) = (column,row). Confusing these conventions can reverse coordinates in placement measurements. Always inspect shape and dtype before processing.

OpenCV cv2.subtract on uint8 saturates negative results at zero. NumPy subtraction on uint8 arrays may wrap around modulo 256 depending on the operation and dtype. Convert to a signed or floating type before arithmetic when negative differences must be preserved.

Task 3 – Image Representation Analysis

Recommendation and Network Features

Lab is recommended as the main colour representation for subtle flux or solder-colour differences, supported by grayscale for edges and HSV for mask/pad segmentation where useful. No single colour space should be assumed to solve every defect. Region-specific features and controlled illumination provide stronger evidence.

Pretrained ResNet or DINOv2 embeddings can encode texture and appearance beyond hand-designed colour thresholds. However, their features are less directly interpretable than a ΔE value or local intensity statistic. For a customer audit, the system should retain a visual ROI and explicit measurements alongside any model score.

Report actual segmentation and classification results only after running the selected code on labelled PCB images. If a real sample set is unavailable, provide the code and a clearly marked synthetic demonstration rather than claiming measured accuracy.

Representation	Useful information	Weakness	Suggested use
Grayscale	Intensity and edges	Loses hue	Edges and local contrast
HSV	Hue, saturation, value	Hue wrap and glare	Mask/pad segmentation
Lab	Lightness and colour axes	Needs correct conversion	Colour difference and flux

Task 4 – Connectivity Analysis

Pixel Connectivity and Defect Regions

After thresholding a solder region, connected-component labelling groups foreground pixels into objects. The chosen connectivity rule determines whether diagonal pixels belong to the same component. This matters when a thin solder bridge runs diagonally between pads or when two separate pads touch only at a corner.

The customer rejects a board with more than two solder balls, any ball larger than 0.3 mm, or any bridge joining neighbouring pads. The pad gap is 0.25 mm. Pixel grouping therefore affects both object count and the decision that two pads have been joined.

The three candidates are 4-connectivity, 8-connectivity and mixed connectivity (m-connectivity). They should be evaluated on synthetic patterns and thresholded board crops, with manual reference labels.

Task 4 – Connectivity Analysis

A – 4-Connectivity

Under 4-connectivity, a pixel is connected only to its north, south, east and west neighbours. In OpenCV, cv2.connectedComponentsWithStats(binary, connectivity=4) labels components using this neighbourhood.

This rule avoids merging objects that touch only diagonally. It can, however, split a thin diagonal bridge into multiple components if no horizontal or vertical pixel path exists. A split bridge may be missed when the algorithm expects one connected region to join two pad areas.

4-connectivity is useful when diagonal contact should not imply physical connection, but it may under-connect diagonal solder structures. A bridge detector should also examine the geometric path between adjacent pads rather than relying on component count alone.

Task 4 – Connectivity Analysis

B – 8-Connectivity

Under 8-connectivity, diagonal neighbours are included in addition to the four orthogonal neighbours. This tends to keep diagonal lines and thin diagonal bridges connected. It is simple to apply using cv2.connectedComponentsWithStats(binary, connectivity=8).

The disadvantage is over-merging. Two pads or solder balls that touch only at a diagonal corner can be labelled as one component, causing an incorrect bridge decision or an inaccurate ball count. Small noise pixels can also connect otherwise separate regions.

8-connectivity is a useful first pass for candidate extraction, but it should be paired with shape, width, pad-location and distance checks. It is not sufficient by itself to prove that a physical electrical bridge exists.

Task 4 – Connectivity Analysis

C – Mixed Connectivity

Mixed or m-connectivity permits diagonal connections only when the diagonal pair does not share a common 4-neighbour that would create an ambiguous connection. This rule reduces some of the ambiguity between 4- and 8-connectivity.

OpenCV does not provide a direct m-connectivity flag in connectedComponentsWithStats, so a custom labelling implementation is required. The implementation should be tested against small binary patterns where the expected labels are known.

A custom implementation is more complex and may be slower than the built-in routines. It is justified only if the test patterns show that it resolves important diagonal cases better than a combined 8-connectivity and geometric validation strategy.

Task 4 – Connectivity Analysis

Distance Measures and Calibration

Euclidean distance is $D_e = \sqrt{(\Delta x^2 + \Delta y^2)}$, city-block distance is $D_4 = |\Delta x| + |\Delta y|$, and chessboard distance is $D_8 = \max(|\Delta x|, |\Delta y|)$. They measure separation differently. Euclidean distance corresponds to straight-line geometry and is usually the most meaningful for physical ball diameter and pad gap.

Convert pixels to millimetres using a calibrated scale. If the calibrated scale is s mm/pixel, a measured diameter dpx corresponds to $d_{mm} = s \times d_{px}$. The scale should be derived from a calibration target at the PCB plane and checked across the field of view.

The 0.3 mm ball limit and 0.25 mm pad gap must be evaluated with uncertainty. If a measured diameter is close to 0.3 mm, include calibration error, threshold-boundary uncertainty and pixel quantization rather than reporting a falsely precise value.

Task 4 – Connectivity Analysis

Validation Patterns and Error Analysis

Create at least five synthetic patterns: separated circular blobs, orthogonally touching blobs, diagonally touching blobs, a thin diagonal bridge and a noisy bridge with isolated pixels. Add at least ten thresholded PCB crops representing real pad geometries. Save the binary input and labelled output for each method.

Compare automatic solder-ball counts and bridge decisions with manually verified references. Record true positives, false positives, false negatives and execution time. Also measure ball diameter error in millimetres and the number of cases in which two distinct pads are merged.

Noise sensitivity can be reduced by morphological filtering and minimum-area constraints, but filtering must not remove a real 0.25 mm defect. Tune the kernel on representative data and preserve the unfiltered mask for audit.

Task 4 – Connectivity Analysis

Recommendation and Segmentation Models

Recommended strategy: use 8-connectivity to generate candidate regions, then verify candidate bridges using pad geometry, minimum width and the path across the known pad gap. Use Euclidean distance for physical measurements. Where diagonal ambiguity remains important, compare custom m-connectivity with the hybrid method on the same test set.

Instance segmentation models such as YOLO-seg or SAM 2 can produce object masks that are less dependent on connected pixel topology. They require representative training or prompt validation, compute resources and careful assessment of touching objects. Their masks must still be calibrated before physical measurements are trusted.

For the specified four-core CPU without a GPU, classical connected components are likely to be simpler and faster. A segmentation model should be selected only if it demonstrates a meaningful reduction in missed bridges or merged objects under the actual deployment conditions.

Connectivity	Diagonal pixels	Typical risk	Use
4-connectivity	Not connected	Splits diagonal bridge	Orthogonal structures
8-connectivity	Connected	Merges diagonal pads	Candidate extraction
m-connectivity	Conditional connection	Custom implementation	Ambiguous patterns

Task 5 – Image Acquisition Analysis

Acquisition Requirements and Motion

The PCB travels at 0.4 m/s. The image acquisition system must freeze motion, avoid geometric distortion and maintain stable colour despite 100 Hz lighting flicker and daylight variation from approximately 300 to 8,000 lux. Shiny solder joints make reflections an important part of the acquisition design.

The three arrangements are area-scan rolling-shutter CMOS, area-scan global-shutter CMOS with strobe and line-scan imaging synchronized to a belt encoder. They differ in exposure timing and how motion is represented in the image.

At the specified rate of two boards per second, acquisition must support a board arrival every 0.5 seconds. A global-shutter camera with a short LED flash is a strong candidate because all pixels are exposed at substantially the same time.

Task 5 – Image Acquisition Analysis

A – Rolling-Shutter Area Scan

A rolling-shutter sensor exposes different image rows at different times. During conveyor motion, the top and bottom of the board may be captured at slightly different positions. This creates skew or geometric distortion even if the exposure of each row is short.

Manual exposure control through cv2.VideoCapture can help reduce blur, but a USB camera may not expose the timing and trigger controls required for deterministic industrial acquisition. Ambient tube lights and daylight can also produce brightness bands or frame-to-frame changes.

Rolling shutter may be acceptable for low-cost monitoring of stationary or slow-moving objects. It is less suitable for precision measurement on a moving conveyor unless the readout time is short and the distortion is calibrated and controlled.

Task 5 – Image Acquisition Analysis

B – Global-Shutter with Strobe

A global-shutter sensor exposes all pixels at approximately the same time. A short LED strobe triggered by a board-presence sensor or encoder freezes motion and reduces dependence on ambient light. The strobe should be synchronized with the exposure and set to a repeatable intensity.

Global shutter avoids the row-time displacement associated with rolling shutter. The camera should use a hardware trigger, fixed gain, fixed exposure and an enclosed or shielded lighting region. A diffuse LED dome can reduce hard reflections from shiny solder, while a controlled low-angle source may reveal surface shape.

This arrangement is recommended for the PCB line because it provides a good balance of motion performance, colour imaging, integration complexity and cost. The final choice still depends on the selected camera's trigger, frame-rate and sensor specifications.

Task 5 – Image Acquisition Analysis

C – Line-Scan Camera

A line-scan camera captures one row at a time. Conveyor movement builds the two-dimensional image by stacking successive rows. The line rate must be synchronized with belt displacement using an encoder; otherwise objects stretch or compress when belt speed changes.

For a desired along-belt sampling of p mm per line and speed v mm/s, the required line rate is fline = v / p. If the desired sampling is approximately 0.10 mm per line at 400 mm/s, the line rate is 4,000 lines/s. The actual rate depends on the selected resolution and optical magnification.

Line scan is useful for long continuous objects and high-speed inspection, but requires precise mechanical and encoder integration. For individual PCBs within a 200 mm field of view, a triggered area-scan camera is likely simpler.

Task 5 – Image Acquisition Analysis

Motion Blur Calculations

Motion blur distance is $b = v \times t$, where v is conveyor speed and t is exposure time. At 0.4 m/s, an exposure of 0.05 ms gives $0.4 \times 0.00005 = 0.00002$ m = 0.02 mm. An exposure of 0.5 ms gives 0.20 mm. An exposure of 2 ms gives 0.80 mm.

The 0.25 mm defect is much smaller than the 0.80 mm blur produced by a 2 ms exposure. Such blur can smear a solder ball or bridge and compromise the placement measurement. A 0.05 ms exposure limits calculated motion blur to 0.02 mm, before considering lens blur, vibration and sensor sampling.

A strobe can provide a short effective illumination pulse even when the camera's frame period is longer. The pulse duration, trigger jitter and light output must be validated together.

Task 5 – Image Acquisition Analysis

Sensor Conversion and Colour

Incident photons generate charge in the photosensitive sensor elements. The accumulated charge is converted to a voltage, conditioned by analogue circuitry, sampled by an analogue-to-digital converter and encoded as a digital pixel value. Gain, full-well capacity, read noise and ADC bit depth affect the quality of the resulting image.

CMOS sensors integrate photodiodes and readout circuitry at the pixel or column level and are widely available in industrial global-shutter cameras. CCD sensors can provide uniform image quality but are less common in modern high-speed industrial systems. A suitable global-shutter CMOS camera is recommended for availability and integration.

A Bayer colour filter array estimates colour from neighbouring photosites through demosaicing. For highly accurate solder colour, compare Bayer colour imaging with monochrome capture and sequential coloured strobes. The latter can improve controlled channel measurements but increases acquisition complexity.

Task 5 – Image Acquisition Analysis

Recommendation and Failure Conditions

Choose a global-shutter CMOS area-scan camera with short synchronized LED strobe, fixed exposure and an enclosure or light shield. Use a diffuse dome or carefully arranged diffuse illumination to reduce specular variation. Keep the encoder or board trigger available for consistent acquisition timing.

If the system uses a 2 ms exposure, calculated blur is 0.8 mm and the smallest 0.25 mm defect may be obscured. If line scan is used without synchronization and belt speed varies by 10%, the image scale along the travel direction also varies by approximately 10%. Corrective measures include shorter exposure, strobe triggering and encoder-based line timing.

3-D laser triangulation or structured light may estimate solder height or volume, but it adds calibration, occlusion, data-rate and cost challenges. It can be considered as a later upgrade for solder-volume inspection rather than the first deployment within the stated budget.

Arrangement	Motion behaviour	Lighting control	Assessment
Rolling shutter	Row skew possible	Ambient-sensitive	Low-cost monitoring
Global shutter + strobe	Common exposure time	High control	Recommended
Line scan	Encoder-built image	Needs synchronization	Continuous surfaces

Task 6 – Sampling Analysis

Spatial Sampling and Field of View

Spatial sampling determines how many pixels represent a physical distance. The camera must cover a 250 mm field across the belt and a 200 mm field along the belt. The smallest specified defect is 0.25 mm, and placement accuracy must be ±0.1 mm.

The three candidate resolutions are 1280 × 1024 (approximately 1.3 MP), 2448 × 2048 (approximately 5 MP) and 4000 × 3000 (12 MP). The relevant scale depends on which image dimension is mapped to the 250 mm field width.

Assuming the horizontal image dimension spans 250 mm, the scale is FOV / pixel count. This assumption should be confirmed against the sensor orientation and lens framing. Calibration at the PCB plane is required to convert image coordinates into millimetres.

Task 6 – Sampling Analysis

Resolution and Millimetres per Pixel

For 1280 pixels across 250 mm, scale = 250/1280 = 0.1953 mm/pixel. A 0.25 mm defect spans about 1.28 pixels. This is below the preferred three-to-four-pixel practical sampling and is marginal even for the two-sample Nyquist minimum.

For 2448 pixels across 250 mm, scale = 0.1021 mm/pixel. A 0.25 mm defect spans about 2.45 pixels. This is above two samples but below the preferred three-to-four pixels, so segmentation and measurement may remain sensitive to blur and threshold choice.

For 4000 pixels across 250 mm, scale = 0.0625 mm/pixel. A 0.25 mm defect spans 4 pixels. This meets the preferred practical sampling target, but the larger image increases data volume and processing demand.

Task 6 – Sampling Analysis

Nyquist and Defect Detection

The Nyquist principle requires at least two samples across the smallest spatial feature to represent it without fundamental undersampling. In practical inspection, three or four pixels across a defect are preferable because real defects have irregular boundaries, optical blur and threshold uncertainty.

The 1.3 MP candidate provides only about 1.28 pixels across a 0.25 mm defect and should not be selected for reliable detection of that defect size. The 5 MP candidate gives about 2.45 pixels and may be acceptable only with strong optics, short exposure and validated defect recall. The 12 MP candidate gives approximately four pixels and is the strongest sampling option.

Sampling is not the only determinant of detectability. Lens modulation transfer, focus, sensor noise, motion blur, lighting and compression can all reduce effective detail. A high pixel count cannot compensate for poor optical contrast.

Task 6 – Sampling Analysis

Image Data Rate and Storage

At two boards per second, raw image data rate is image bytes × 2. For an 8-bit BGR image, each pixel uses three bytes. The 1280 × 1024 image is about 3.93 MB, or 7.86 MB/s at two boards/s. The 2448 × 2048 image is about 15.04 MB, or 30.08 MB/s. The 4000 × 3000 image is 36 MB, or 72 MB/s.

Seven-day storage at continuous operation is calculated as daily data × 7. If the line operates for ten hours per day, 72,000 boards are imaged per day. The uncompressed BGR storage is approximately 283 GB/day for 1.3 MP, 1.08 TB/day for 5 MP and 2.59 TB/day for 12 MP. These figures exceed the 2 TB device over seven days unless images are compressed or the duty cycle is lower.

Storage planning must include metadata, file-system overhead, rejected-image retention and backup policy. Lossless compression may reduce storage but the actual ratio depends on image content. JPEG is not preferred for subtle solder defects because lossy compression can alter small intensity differences.

Task 6 – Sampling Analysis

Frame Rate and Board Coverage

The along-belt field of view is 200 mm and the maximum board length is 180 mm. At 0.4 m/s, the time for a board to travel through the 200 mm field is 0.2/0.4 = 0.5 seconds. To guarantee at least one complete frame for a free-running camera, the frame period must not exceed the available full-board capture interval; a trigger-based capture is preferable.

The board arrival rate is two boards per second, corresponding to 0.5 seconds between boards. The camera should support at least two triggered full-resolution frames per second with sufficient transfer and processing headroom. If boards can overlap within the field or arrive irregularly, use a presence sensor and encoder rather than relying on a fixed free-running frame rate.

The camera-to-gate travel time is 1.5 seconds. The vision decision, communication, PLC logic and gate actuation must all fit within this time.

Task 6 – Sampling Analysis

Down-Sampling and Aliasing

Use cv2.resize with INTER_NEAREST and INTER_AREA to compare down-sampling methods. Nearest-neighbour sampling selects one source pixel and can produce jagged edges and aliasing. INTER_AREA averages contributions over the destination pixel area and is generally more suitable for shrinking images.

Create a fixed high-resolution reference capture and resize it to each candidate resolution. Measure defect diameter and placement coordinate before and after resizing. Compare edge profiles, connected-component counts and defect recall. Keep the original image as the audit record.

Upscaling a low-resolution image does not create new optical evidence. Deep super-resolution methods such as Real-ESRGAN can generate plausible details, but generated pixels are not reliable proof that a 0.25 mm defect existed. Do not use synthesized detail for metrology or customer-audited acceptance.

Task 6 – Sampling Analysis

Resolution Recommendation

The 12 MP option provides approximately four pixels across a 0.25 mm defect under the stated 250 mm horizontal field. It offers the best sampling margin but its raw data volume and CPU processing load are high. The 5 MP option is a compromise but offers only about 2.45 pixels across the defect.

Because the customer allows at most 1% escaped defective boards, selecting the 1.3 MP option solely to save cost is not defensible without evidence that the smallest defect is still detected. The preferred approach is a 5 MP global-shutter system only if optics, exposure and testing demonstrate the required recall; otherwise choose a cropped or higher-resolution acquisition strategy within the available budget.

Before final purchase, run a measured benchmark on the four-core edge PC using full-resolution images, lossless storage and all pipeline stages. If 12 MP cannot sustain two boards per second, use carefully defined regions of interest, efficient data transfer and a tested image-retention policy without discarding original evidence.

Resolution	Scale (mm/px)	0.25 mm defect (px)	Assessment
1280 px	0.1953	1.28	Insufficient margin
2448 px	0.1021	2.45	Borderline; validate
4000 px	0.0625	4.00	Best sampling; heavy data

Task 7 – Quantization Analysis

Quantization and Solder Contrast

Quantization maps continuous sensor output to a finite number of digital levels. The question states that insufficient solder differs from a good joint by only 8–12 grey levels in an 8-bit image. The quantization step must be small enough to preserve this difference after noise and illumination variation.

An 8-bit image contains 256 intensity levels. A 12-bit raw image contains 4096 levels. Reduced 5-bit and 6-bit images contain 32 and 64 levels respectively. Higher bit depth can preserve subtle tonal differences, but only if the sensor noise, exposure and analogue electronics support the additional information.

The three candidates are 8-bit, 12-bit raw and reduced 5–6 bit. Evaluate them on the same solder ROIs and under identical exposure and illumination conditions.

Task 7 – Quantization Analysis

A – 8-Bit Image

An 8-bit unsigned image uses values from 0 to 255. It is directly supported by common OpenCV operations and is relatively efficient to store and process. The stated 8–12 level difference corresponds to several quantization steps and may be visible when lighting is stable.

However, the actual separation can be reduced by sensor noise, compression, glare, shadows and exposure drift. Histogram equalisation may improve visual contrast but does not increase the original bit depth or restore clipped highlights.

8-bit lossless storage is a practical baseline. Keep camera gain and exposure fixed and verify that good and insufficient solder distributions remain separated under the worst expected lighting condition.

Task 7 – Quantization Analysis

B – 12-Bit Raw

A 12-bit sensor output provides 4096 possible codes. If stored in a 16-bit container, each channel is typically represented using two bytes even though only 12 bits carry signal information. The extra levels can preserve subtle intensity differences and provide more headroom for later normalization.

Do not convert raw 12-bit data to 8-bit by an arbitrary cast. Use a controlled scaling method such as normalization or convertScaleAbs for display while retaining the original raw data for measurement. A display image is not a substitute for the raw acquisition.

12-bit acquisition is most useful when the sensor has sufficient dynamic range and noise performance. It cannot overcome photon noise or a saturated specular highlight, so lighting and exposure remain essential.

Task 7 – Quantization Analysis

C – Reduced 5- and 6-Bit

Reducing an 8-bit image to 6 bits creates 64 levels, while 5 bits creates 32 levels. A simple quantizer maps groups of input values to one output value. This saves storage only if the file format or packed representation actually stores fewer bits; writing the values into a uint8 array does not automatically reduce file size.

A 5-bit representation has a nominal step of about eight 8-bit levels when uniformly mapped across the range. The specified 8–12 grey-level difference may therefore be reduced to one or two quantization intervals, or lost entirely depending on where the values fall relative to quantization boundaries.

Reduced bit depth also causes false contouring in smooth gradients and can change colour difference measurements. It is not recommended for the master audit image.

Task 7 – Quantization Analysis

Contrast and Detection Evaluation

Select at least 30 solder joints with good and insufficient-solder labels. Capture them at two lighting levels. For each bit depth, report mean intensity, standard deviation, contrast between classes, overlap of distributions and defect detection rate.

Use a fixed decision threshold selected on a training subset, then evaluate on a held-out test subset. Re-selecting the threshold separately for each test image would make the comparison unfair. Include false alarms and missed defects as well as overall accuracy.

An 8–12 level signal difference is vulnerable to coarse quantization and illumination drift. The preferred acquisition is 12-bit raw where supported, with an 8-bit normalized derivative for fast display or processing if the measured edge-PC performance requires it.

Task 7 – Quantization Analysis

Storage Trade-Off

For a 2448 × 2048 image, 8-bit grayscale requires about 5.01 MB uncompressed, 8-bit BGR about 15.04 MB, and 16-bit container storage for 12-bit BGR about 30.08 MB. Actual camera formats and packing may differ. Multiply per-board size by 72,000 boards/day for a ten-hour, two-board/s shift.

The 2 TB storage device cannot hold seven days of all uncompressed 12-bit BGR images at the 5 MP resolution. A practical design may store raw high-bit-depth images for rejected and uncertain boards, while retaining a lossless compressed image or verified evidence package for all boards, subject to customer approval.

Any selective retention policy must still satisfy the stated requirement that every board image is kept for seven days. Therefore, compression, storage expansion or reduced image payload must be explicitly approved; do not silently discard normal-board images.

Task 7 – Quantization Analysis

HDR, INT8 and Recommendation

Multi-exposure HDR combines exposures to preserve detail across bright and dark regions. On a moving conveyor, sequential exposures can misregister, so a synchronized strobe or single-frame HDR sensor is preferable. HDR addresses scene dynamic range, not the number of bits used by a neural network.

INT8 model quantization reduces numerical precision in neural-network weights and activations to improve inference speed and memory use. It is separate from camera image bit depth. INT8 deployment should be validated for changes in defect recall and false acceptance.

Recommendation: acquire 12-bit raw if the selected camera and budget support it, preserve the original data, and create a calibrated 8-bit processing view. Do not use 5-bit or 6-bit data as the only audit evidence because the subtle insufficient-solder contrast may be lost.

Depth	Levels	Potential benefit	Risk
5 bit	32	Small representation	May erase 8–12 level contrast
6 bit	64	Moderate storage reduction	Coarse transitions
8 bit	256	Standard processing	Limited dynamic range
12 bit	4096	More tonal headroom	Larger storage and processing

Task 8 – Image Formation Analysis

Geometric and Photometric Models

The geometric model describes where a scene point appears in the image. In a pinhole model, $x = fX/Z$ and $y = fY/Z$, where f is focal length and Z is distance from the camera. The photometric model describes pixel brightness as $f(x,y) = i(x,y) \times r(x,y)$, where i is illumination and r is surface reflectance.

PCB placement measurement depends on geometric stability. Solder quality judgement depends on photometric stability. A camera may produce a sharp image but still provide poor measurements if perspective changes with component height or poor colour decisions if the illumination varies.

The three configurations are ambient light with a standard lens, LED dome with a calibrated standard lens, and LED dome with a telecentric lens. Each must be compared for measurement accuracy, illumination uniformity, glare and cost.

Task 8 – Image Formation Analysis

A – Ambient Light and Standard Lens

Ambient illumination from overhead tube lights and daylight changes with time and weather. The 100 Hz flicker can create frame-to-frame brightness changes, while the loading-bay door produces spatially and temporally varying light. Reflectance-based thresholds therefore become unstable.

A standard lens produces perspective projection. Components above the PCB plane are closer to the camera and may appear larger than features at the board plane. A 12 mm tall capacitor can therefore have a different apparent scale from a flat pad.

This arrangement is inexpensive but does not provide sufficient control for ±0.1 mm placement measurement and subtle solder contrast without additional calibration and lighting control.

Task 8 – Image Formation Analysis

B – LED Dome and Calibrated Standard Lens

An LED dome provides more uniform diffuse illumination and can reduce harsh reflections from shiny solder. A fixed-current LED source with a controlled trigger makes illumination repeatable. The camera should be shielded from ambient daylight and the exposure and gain should be locked.

Calibrate lens distortion and image scale using a checkerboard or known calibration target. OpenCV functions include cv2.findChessboardCorners and cv2.calibrateCamera. Use the calibration matrix and distortion coefficients to undistort the image before measuring component coordinates.

Calibration should be performed at the PCB plane and verified across the field of view. A standard lens still exhibits perspective changes with height, so height-dependent measurement error must be quantified.

Task 8 – Image Formation Analysis

C – LED Dome and Telecentric Lens

A telecentric lens is designed so that chief rays are nearly parallel in object space. Apparent object size changes much less with working distance than with a conventional perspective lens. This is valuable for precision dimensional measurement when components have different heights.

Telecentric optics can reduce scale variation caused by a component being 12 mm above the board plane. They are typically more expensive, have limited field of view and may require a larger physical lens. The imaging budget is ₹2,00,000 for camera, lens and lighting, so feasibility must be confirmed using vendor quotations.

A telecentric lens does not automatically solve solder glare or colour variation. It should still be paired with controlled illumination and calibration.

Task 8 – Image Formation Analysis

Focal Length and Perspective Calculation

The question asks for a focal-length calculation and requires the sensor width to be stated. Assume a sensor width of 8.8 mm and a 250 mm horizontal field of view at a 600 mm working distance. From similar triangles, f = (sensor width × working distance) / FOV = (8.8 × 600) / 250 = 21.12 mm.

This is an approximate thin-lens field-of-view calculation. The actual lens selection must account for sensor active area, working distance definition, lens distortion, focus range and mechanical clearances. Confirm the field of view using the manufacturer's lens calculator and a calibration target.

For a perspective camera, image scale is inversely proportional to object distance. If the board plane is 600 mm from the camera and the top of a component is 12 mm closer, the scale ratio is 600/(600−12) ≈ 1.0204, or about 2.04% larger at the component top. The actual sign depends on the working-distance convention.

Task 8 – Image Formation Analysis

Illumination Correction and Reflectance

Flat-field correction estimates the illumination field using a uniform white reference. A corrected image can be formed by dividing the captured image by the normalized illumination reference, with safeguards against division by zero. This reduces slowly varying brightness across the field.

Homomorphic filtering uses log f = log i + log r to separate slowly varying illumination from reflectance detail. It can improve local contrast but may alter colour and fine texture if filter parameters are poorly selected.

Compare defect detection before and after correction using the same thresholds or a documented recalibration procedure. Show illumination maps and report non-uniformity, such as the coefficient of variation across a uniform target.

Task 8 – Image Formation Analysis

Recommendation and Limits

The LED dome with a calibrated standard lens is the recommended baseline because it controls illumination and supports geometric calibration at lower cost than a telecentric lens. If the measured height-induced error exceeds the ±0.1 mm tolerance, a telecentric lens or a mechanical height constraint should be considered.

Daylight should be blocked using an enclosure. Use a stable LED driver, fixed exposure and periodic reference checks. Store calibration images, calibration date, lens settings and the measured residual error.

Monocular depth-estimation networks such as Depth Anything may provide qualitative depth cues but should not be treated as a metrology instrument without calibration and validation. Compare estimated heights against known gauge blocks or calibrated measurements before considering deployment.

Configuration	Geometry	Illumination	Assessment
Ambient + standard lens	Perspective	Uncontrolled	Weak baseline
LED dome + calibrated lens	Perspective corrected	Controlled diffuse light	Recommended baseline
LED dome + telecentric	Near-orthographic	Controlled diffuse light	Best metrology; higher cost

Task 9 – Integrated System and Performance Evaluation

Integrated Design Requirements

Task 9 combines the decisions from Tasks 1–8 into a complete imaging front-end. The system must stay within the ₹2,00,000 imaging budget, inspect two boards per second, make a decision within 1.5 seconds, detect 0.25 mm defects and retain every board image for seven days.

The three required configurations are Budget, Balanced and Premium. A fourth smart-camera option is added for comparison. Cost estimates must be replaced by current vendor quotations before procurement; the question explicitly requests real vendor prices and citations, and no price should be invented.

The selected architecture should be evaluated on the same labelled dataset and under the same lighting, board shift and defect conditions. This ensures that differences are caused by the configuration rather than by different test data.

Task 9 – Integrated System and Performance Evaluation

Configuration A – Budget

The budget configuration uses a 1.3 MP USB rolling-shutter camera, ambient illumination, 8-bit JPEG storage and a low-plus-mid-level pipeline. It has the lowest initial cost and simple integration.

Its major weaknesses are insufficient sampling for a 0.25 mm defect across the 250 mm field, motion distortion, ambient-light variation and lossy compression. The resolution calculation gives about 1.28 pixels across the smallest defect, below the preferred sampling requirement.

This option may be useful for operator assistance or coarse presence checks, but it should not be selected for final customer-audited defect acceptance without strong contrary test evidence.

Task 9 – Integrated System and Performance Evaluation

Configuration B – Balanced

The balanced configuration uses a 5 MP global-shutter GigE camera, LED dome with strobe, calibrated standard lens, 8-bit lossless storage and a three-level pipeline. It provides controlled acquisition and a practical integration path for the four-core edge PC.

At 2448 pixels across 250 mm, the scale is about 0.1021 mm/pixel and a 0.25 mm defect spans approximately 2.45 pixels. This is better than the budget option but remains below the preferred three-to-four-pixel target. It therefore requires careful optics, short exposure and defect-specific validation.

This is the leading candidate if actual tests demonstrate acceptable defect recall and latency. It should be paired with a clear reject policy for uncertain cases and retained raw or lossless image evidence.

Task 9 – Integrated System and Performance Evaluation

Configuration C – Premium

The premium configuration uses a 12 MP area-scan or line-scan camera, telecentric lens and 12-bit storage with a three-level pipeline. It provides the strongest spatial and intensity sampling among the three options.

At 4000 pixels across 250 mm, the scale is 0.0625 mm/pixel and the 0.25 mm defect spans about four pixels. However, 12 MP BGR at two boards per second produces approximately 72 MB/s uncompressed and around 2.59 TB for a ten-hour day, before seven-day retention.

The premium option may exceed both the ₹2,00,000 budget and the 2 TB storage limit. It should be considered only if a verified cost and storage design is available.

Task 9 – Integrated System and Performance Evaluation

Fourth Option – Smart Camera

A smart camera with an onboard AI accelerator can combine image acquisition and inference in one industrial device. It may simplify wiring and reduce the load on the edge PC. A compact object detector could identify components and solder defect candidates.

The disadvantages include vendor lock-in, model deployment constraints, limited flexibility for custom measurement and the need to validate the accelerator's actual latency. The smart camera must still provide sufficient spatial sampling, controlled illumination and audit image export.

Compare smart-camera cost, total system cost, storage, robustness, maintenance and retraining effort. It is not automatically cheaper than a GigE camera plus edge PC.

Task 9 – Integrated System and Performance Evaluation

End-to-End Flow and Time Budget

Recommended block flow: PCB and conveyor → trigger/encoder → LED dome and strobe → lens and global-shutter sensor → sampling and quantization → raw image → flat-field correction and undistortion → board alignment → component and solder segmentation → feature extraction → rule/classifier decision → PLC communication → reject gate.

A possible engineering time budget within 1.5 seconds is: trigger and acquisition 50 ms; transfer 100 ms; preprocessing and alignment 150 ms; segmentation and features 300 ms; classification and rules 100 ms; logging 100 ms; PLC and gate reserve 200 ms; remaining margin 500 ms. These are design allocations, not measured timings.

Measure each stage on the actual four-core PC. The final decision must be timestamped and linked to the board's encoder position so that the correct board is rejected.

Task 9 – Integrated System and Performance Evaluation

Other Applications and Final Design

Retinal fundus screening uses visible-light colour images and emphasizes vascular and lesion contrast. Document scanning uses reflected light, high spatial resolution and OCR-oriented text geometry. Drone crop monitoring may use visible and near-infrared bands, large-area imaging and changing outdoor illumination. These differ from PCB inspection in spectral band, sampling, quantization and image-formation assumptions.

Recommended baseline: balanced global-shutter 5 MP GigE camera, LED dome/strobe, calibrated standard lens, lossless image retention and a three-level OpenCV pipeline. This recommendation is conditional on measured defect recall, throughput and seven-day storage compliance. If 5 MP sampling fails the 0.25 mm defect test, redesign the optical field or move to a higher-resolution solution with a verified data-retention plan.

Before commissioning, obtain vendor quotations, run the full labelled test set, test the 1% escape limit with sufficient sample size, validate gate timing, and obtain customer approval for the image-storage format and retention architecture.

Configuration	Core components	Main advantage	Main concern
Budget	1.3 MP rolling shutter	Low cost	Sampling and lighting
Balanced	5 MP global shutter + dome	Practical balance	2.45 px per 0.25 mm defect
Premium	12 MP + telecentric + 12 bit	Strongest detail	Cost, CPU and storage
Smart camera	Onboard AI accelerator	Integrated inference	Vendor lock-in and audit export

Task 10 – Front-End Adaptation Challenge

Changed Requirements and Invalid Assumptions

The new customer requires detection of 0.1 mm hairline cracks and solder balls under conformal coating that fluoresce under UV illumination. Conveyor speed doubles from 0.4 m/s to 0.8 m/s, throughput increases from two to four boards per second, daylight becomes stronger because the loading-bay door remains open, and the decision time falls to 0.75 seconds.

The original assumption that all relevant defects are visible under normal white light is no longer valid. The original 0.25 mm minimum defect is replaced by 0.1 mm. The original exposure and frame-rate assumptions must be revisited because motion doubles. The original lighting stability assumption is invalid because stronger daylight enters the inspection area.

The same four-core edge PC, 16 GB RAM, no GPU and ₹2,00,000 budget must be retained. The adaptation must therefore balance UV sensitivity, spatial resolution, acquisition speed, CPU load and storage.

Task 10 – Front-End Adaptation Challenge

Spatial Sampling for 0.1 mm Defects

Using the preferred target of four pixels across a 0.1 mm defect requires a scale of 0.1/4 = 0.025 mm/pixel. Across a 250 mm field, this requires 250/0.025 = 10,000 pixels horizontally. Even the minimum two-sample Nyquist criterion requires 0.05 mm/pixel, or 5,000 pixels across the field.

A 12 MP camera with 4000 pixels across 250 mm provides 0.0625 mm/pixel, so a 0.1 mm defect spans only 1.6 pixels. It is insufficient for reliable practical detection at the original field width. A higher-resolution sensor, a narrower region of interest, or multiple views may be needed.

An ROI strategy can increase effective sampling only if the optics or sensor readout provide more pixels per millimetre in that ROI. Cropping an already captured low-resolution image does not add spatial information.

Task 10 – Front-End Adaptation Challenge

Motion Blur and Acquisition Rate

At 0.8 m/s, blur b = v × t. For 0.05 ms exposure, blur = 0.8 × 0.00005 = 0.00004 m = 0.04 mm. For 0.5 ms, blur = 0.40 mm. For 2 ms, blur = 1.60 mm. The latter two exposures are much larger than the 0.1 mm defect.

A short strobe pulse is therefore essential. The camera must also support four boards per second with enough time for transfer and processing. The board arrival interval is 0.25 seconds. The new gate distance is 0.75 m; at 0.8 m/s, travel time is 0.75/0.8 = 0.9375 seconds, so the stated 0.75-second decision requirement should be treated as the stricter system deadline.

For line scan, required line rate is conveyor speed divided by desired along-belt pixel spacing. At 0.8 m/s and 0.025 mm per line, fline = 800/0.025 = 32,000 lines/s. The selected sensor and encoder must sustain this rate.

Task 10 – Front-End Adaptation Challenge

Strategy A – Retain Camera with ROI

Strategy A retains the Task 9 camera and lighting, shortens exposure and crops to the board ROI. It is the least expensive adaptation and can reduce CPU processing and data transfer when the original image contains large unused margins.

However, ROI processing cannot recover 0.1 mm detail that the sensor did not sample. The original 5 MP and 12 MP examples provide approximately 2.45 and 4 pixels across a 0.25 mm defect, respectively; for a 0.1 mm defect the corresponding values fall to about 0.98 and 1.6 pixels across the same field.

Strategy A also cannot detect defects that are invisible in white light but fluoresce under UV. It is therefore not a complete solution to the new customer requirement.

Task 10 – Front-End Adaptation Challenge

Strategy B – Modify Sampling and Acquisition

Strategy B uses a higher-resolution global-shutter camera, a shorter strobe and an enclosed hood to block daylight. A 10- or 12-bit sensor can preserve subtle intensity variation, while the enclosure stabilizes illumination.

This strategy improves sampling and motion control but still needs a UV-sensitive optical path or UV illumination if the defect only fluoresces under UV. A white-light-only camera configuration cannot provide the missing spectral evidence.

The required horizontal sampling of 5,000 pixels for two samples or 10,000 pixels for four samples across 250 mm may be difficult within the budget. A narrower optical field or multiple ROIs should be assessed.

Task 10 – Front-End Adaptation Challenge

Strategy C – Redesign with White and UV

Strategy C captures white-light and UV-strobe images sequentially or uses two cameras. White light supports component placement and ordinary solder inspection; UV excitation reveals fluorescent cracks and solder-related defects beneath the coating. The inspection area must be enclosed to exclude daylight and protect repeatability.

Sequential capture reduces hardware cost but adds trigger and registration requirements. Two cameras provide parallel acquisition but increase cost, calibration effort and data volume. UV wavelength, filter transmission, sensor sensitivity and coating fluorescence must be selected from measured material response rather than assumed.

Calibration should include a UV reference target and a spatial registration method between white and UV images. Store both channels with linked board identifiers so that an audit can reproduce the combined decision.

Task 10 – Front-End Adaptation Challenge

Final Decision, Learning and Audit

Recommended decision: REDESIGN the sensing front-end using an enclosed, synchronized white-plus-UV illumination arrangement and a global-shutter sensor. Use a narrow ROI or multi-view optics to achieve the required 0.025–0.05 mm/pixel sampling where the critical defect occurs. Keep the classical OpenCV geometry pipeline for placement and use UV-specific intensity/texture features for fluorescent defects.

Anomaly detection or few-shot learning may reduce the number of labelled UV defect examples needed, but it is vulnerable to limited samples, domain shift and false acceptance. Build a normal reference set across boards, coating batches and operating conditions; use held-out lots for validation; log model version and thresholds; and require human review for uncertain cases.

The design is no longer suitable if the defect is smaller than the validated optical resolution, UV fluorescence is weak or inconsistent, the line exceeds the tested four-board/s rate, or the budget cannot support the required sensor and optics. These limits should be stated before production approval.

Strategy	What changes	What remains weak	Decision
A: Retain + ROI	Exposure and processing area	No new UV evidence; sampling limit	Reject as complete solution
B: Higher sampling	Resolution, strobe, enclosure	UV-only defect visibility	Partial modification
C: White + UV redesign	Spectral sensing and enclosure	Cost and registration complexity	Recommended redesign